

All Parametric Survival Models

METIMMOX-1 Economic Evaluation

Ben Geisler

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1 Overview

This technical document presents the regression coefficients for **all parametric survival models** fitted in the METIMMOX-1 economic evaluation. While the primary analysis uses the best-fitting model (selected by AIC), this document provides full transparency by showing coefficients for all distributions tested.

1.1 Model Specification

All models use the following covariate structure (full model with age and sex adjustment):

Overall Survival (OS):

$\text{Surv(OSwk, Death)} \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp}*\text{Rx} + \text{tlr}*\text{Rx} + \text{tmb_braf}*\text{Rx}$

Progression-Free Survival (PFS):

$\text{Surv(PFSwk, Progression)} \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp}*\text{Rx} + \text{tlr}*\text{Rx} + \text{tmb_braf}*\text{Rx}$

1.2 Distributions Tested

Nine parametric distributions were fitted to each outcome:

1. **Exponential** - Constant hazard rate
2. **Weibull (AFT)** - Accelerated failure time parameterization
3. **Weibull (PH)** - Proportional hazards parameterization
4. **Log-logistic** - Allows for non-monotonic hazard
5. **Log-normal** - Log-transformed normal distribution for survival times
6. **Gamma** - Two-parameter distribution, flexible shape
7. **Gompertz** - Hazard increases exponentially with time
8. **Generalized Gamma** - Three-parameter family including Weibull, gamma, and log-normal as special cases
9. **Generalized F** - Four-parameter family, most flexible

2 Model Selection Summary

2.1 Overall Survival (OS) Models

Table 1: Overall Survival model fit comparison (sorted by AIC)

Distribution	AIC	BIC	Delta AIC	Rank	Selected
Gamma	622.56	646.47	0.00	1.0	***
Weibull (AFT)	623.44	647.36	0.88	2.5	
Weibull (PH)	623.44	647.36	0.88	2.5	
Log-logistic	623.68	647.60	1.12	4.0	
Log-normal	624.09	648.01	1.53	5.0	
Generalized Gamma	624.55	650.65	1.99	6.0	
Generalized F	626.56	654.82	4.00	7.0	
Gompertz	633.09	657.01	10.53	8.0	
Exponential	654.14	675.88	31.58	9.0	

2.2 Progression-Free Survival (PFS) Models

Table 2: Progression-Free Survival model fit comparison (sorted by AIC)

Distribution	AIC	BIC	Delta AIC	Rank	Selected
Gamma	453.43	477.35	0.00	1.0	***
Log-normal	453.46	477.38	0.03	2.0	
Weibull (AFT)	454.19	478.11	0.76	3.5	
Weibull (PH)	454.19	478.11	0.76	3.5	
Log-logistic	455.16	479.08	1.73	5.0	
Generalized Gamma	455.28	481.37	1.85	6.0	
Generalized F	457.28	485.55	3.85	7.0	
Gompertz	467.57	491.49	14.14	8.0	
Exponential	483.52	505.26	30.09	9.0	

3 Overall Survival (OS) Model Coefficients

Best-fitting distribution: gamma (AIC: 622.56)

The following tables present the regression coefficients for each fitted OS model. Parameters include:

- **Distribution-specific parameters** (e.g., shape, scale, rate)
- **Age** - Effect per year of age
- **Sex** - Effect of male vs female
- **Rx** - Main treatment effect (experimental vs control)
- **Biomarker:Rx interactions** - Treatment effect modification by biomarker status

3.1 Gamma (Best Fit)

Parameter	Estimate	SE	exp(Est)
shape	1.2264	0.1821	3.4090
rate	-2.8997	0.6161	0.0550
Age (per year)	0.0015	0.0082	1.0015
Sex (male vs female)	0.1442	0.1482	1.1551
RxExperimental arm	-0.2348	0.2852	0.7907
crp1	-0.3621	0.3169	0.6962
tlr1	-0.8044	0.2509	0.4473
tmb_braf1	-0.0890	0.2210	0.9148
RxExperimental arm:crp1	0.0207	0.3913	1.0209
RxExperimental arm:tlr1	0.3953	0.3295	1.4849
RxExperimental arm:tmb_braf1	-0.0946	0.3018	0.9098

3.2 Exponential

Parameter	Estimate	SE	exp(Est)
rate	-4.4248	1.1022	0.0120
Age (per year)	0.0063	0.0156	1.0064
Sex (male vs female)	0.1835	0.2830	1.2014
RxExperimental arm	-0.1916	0.5253	0.8256
crp1	-0.5534	0.6427	0.5750
tlr1	-0.9110	0.4699	0.4021
tmb_braf1	-0.1924	0.4242	0.8250
RxExperimental arm:crp1	0.1371	0.7793	1.1470
RxExperimental arm:tlr1	0.3861	0.6193	1.4712
RxExperimental arm:tmb_braf1	-0.1050	0.5789	0.9003

3.3 Generalized F

Parameter	Estimate	SE	exp(Est)
mu	4.1157	0.6323	61.2949
sigma	-0.6103	0.1367	0.5432
Q	0.5270	0.4282	1.6938
P	-7.1738	65.3299	0.0008
Age (per year)	-0.0014	0.0082	0.9986
Sex (male vs female)	-0.1447	0.1492	0.8653
RxExperimental arm	0.2322	0.2875	1.2613
crp1	0.3559	0.3180	1.4275
tlr1	0.8046	0.2519	2.2358
tmb_braf1	0.0880	0.2243	1.0920
RxExperimental arm:crp1	-0.0153	0.3961	0.9848
RxExperimental arm:tlr1	-0.3924	0.3319	0.6754
RxExperimental arm:tmb_braf1	0.0996	0.3025	1.1047

3.4 Generalized Gamma

Parameter	Estimate	SE	exp(Est)
mu	4.1285	0.6310	62.0816
sigma	-0.6097	0.1365	0.5435
Q	0.5269	0.4277	1.6936
Age (per year)	-0.0016	0.0082	0.9984
Sex (male vs female)	-0.1461	0.1492	0.8641
RxExperimental arm	0.2282	0.2875	1.2563
crp1	0.3510	0.3177	1.4205
tlr1	0.8048	0.2520	2.2363
tmb_braf1	0.0845	0.2243	1.0882
RxExperimental arm:crp1	-0.0088	0.3959	0.9912
RxExperimental arm:tlr1	-0.3942	0.3320	0.6742
RxExperimental arm:tmb_braf1	0.1041	0.3025	1.1097

3.5 Gompertz

Parameter	Estimate	SE	exp(Est)
shape	0.0137	0.0026	1.0138
rate	-5.5282	1.1991	0.0040
Age (per year)	0.0126	0.0170	1.0127
Sex (male vs female)	0.3678	0.2983	1.4446
RxExperimental arm	-0.3428	0.5268	0.7098
crp1	-1.1291	0.6666	0.3233
tlr1	-1.4740	0.4923	0.2290
tmb_braf1	-0.1956	0.4296	0.8224
RxExperimental arm:crp1	0.4749	0.8020	1.6079
RxExperimental arm:tlr1	0.7020	0.6346	2.0178
RxExperimental arm:tmb_braf1	-0.0821	0.5975	0.9212

3.6 Log-logistic

Parameter	Estimate	SE	exp(Est)
shape	1.0880	0.1125	2.9683
scale	3.6647	0.6313	39.0442
Age (per year)	0.0028	0.0084	1.0028
Sex (male vs female)	-0.1517	0.1501	0.8592
RxExperimental arm	0.2843	0.3371	1.3288
crp1	0.2045	0.3493	1.2269
tlr1	0.8812	0.2840	2.4137
tmb_braf1	0.0838	0.2482	1.0874
RxExperimental arm:crp1	0.1383	0.4134	1.1484
RxExperimental arm:tlr1	-0.4512	0.3539	0.6368
RxExperimental arm:tmb_braf1	0.1421	0.3165	1.1527

3.7 Log-normal

Parameter	Estimate	SE	exp(Est)
meanlog	3.9003	0.6356	49.4154
sdlog	-0.5154	0.0969	0.5973
Age (per year)	-0.0013	0.0083	0.9987
Sex (male vs female)	-0.1283	0.1548	0.8796
RxExperimental arm	0.2676	0.3244	1.3068
crp1	0.3716	0.3345	1.4501
tlr1	0.7722	0.2727	2.1645
tmb_braf1	0.1527	0.2373	1.1650
RxExperimental arm:crp1	0.0144	0.4121	1.0145
RxExperimental arm:tlr1	-0.3129	0.3608	0.7313
RxExperimental arm:tmb_braf1	0.0806	0.3190	1.0839

3.8 Weibull (AFT)

Parameter	Estimate	SE	exp(Est)
shape	0.7281	0.1090	2.0711
scale	4.5338	0.5431	93.1093
Age (per year)	-0.0051	0.0081	0.9949
Sex (male vs female)	-0.1654	0.1429	0.8475
RxExperimental arm	0.2082	0.2536	1.2315
crp1	0.4164	0.3091	1.5164
tlr1	0.7730	0.2295	2.1663
tmb_braf1	0.0673	0.2081	1.0696
RxExperimental arm:crp1	-0.0990	0.3815	0.9058
RxExperimental arm:tlr1	-0.3977	0.3021	0.6719
RxExperimental arm:tmb_braf1	0.0578	0.2888	1.0595

3.9 Weibull (PH)

Parameter	Estimate	SE	exp(Est)
shape	0.7281	0.1090	2.0711
scale	-9.3901	1.5795	0.0001
Age (per year)	0.0105	0.0168	1.0106
Sex (male vs female)	0.3426	0.2988	1.4086
RxExperimental arm	-0.4313	0.5275	0.6497
crp1	-0.8624	0.6443	0.4222
tlr1	-1.6010	0.4921	0.2017
tmb_braf1	-0.1394	0.4303	0.8699
RxExperimental arm:crp1	0.2049	0.7904	1.2275
RxExperimental arm:tlr1	0.8237	0.6317	2.2789
RxExperimental arm:tmb_braf1	-0.1197	0.5978	0.8872

4 Progression-Free Survival (PFS) Model Coefficients

Best-fitting distribution: gamma (AIC: 453.43)

The following tables present the regression coefficients for each fitted PFS model.

4.1 Gamma (Best Fit)

Parameter	Estimate	SE	exp(Est)
shape	1.2423	0.1901	3.4637
rate	-1.8724	0.6325	0.1538
Age (per year)	-0.0008	0.0084	0.9992
Sex (male vs female)	0.1330	0.1653	1.1423
RxExperimental arm	0.4228	0.3072	1.5262
crp1	0.0663	0.4254	1.0685
tlr1	-0.9773	0.2717	0.3763
tmb_braf1	-0.1039	0.2536	0.9013
RxExperimental arm:crp1	-0.6418	0.4911	0.5263
RxExperimental arm:tlr1	-0.2142	0.3486	0.8072
RxExperimental arm:tmb_braf1	-0.3863	0.3409	0.6796

4.2 Exponential

Parameter	Estimate	SE	exp(Est)
rate	-3.0223	1.1468	0.0487
Age (per year)	-0.0026	0.0161	0.9975
Sex (male vs female)	0.1749	0.3161	1.1911
RxExperimental arm	0.3956	0.5700	1.4853
crp1	-0.0452	0.7953	0.9558
tlr1	-1.0805	0.5136	0.3394
tmb_braf1	-0.2308	0.4812	0.7939
RxExperimental arm:crp1	-0.5242	0.9260	0.5920
RxExperimental arm:tlr1	-0.1622	0.6622	0.8503
RxExperimental arm:tmb_braf1	-0.4092	0.6555	0.6642

4.3 Generalized F

Parameter	Estimate	SE	exp(Est)
mu	3.0808	0.6539	21.7760
sigma	-0.5735	0.1450	0.5635
Q	0.2873	0.6748	1.3328
P	-8.5840	66.2147	0.0002
Age (per year)	0.0001	0.0088	1.0001
Sex (male vs female)	-0.1380	0.1651	0.8711
RxExperimental arm	-0.3981	0.3283	0.6716
crp1	0.0162	0.5002	1.0163
tlr1	0.9721	0.2842	2.6435
tmb_braf1	0.0878	0.2655	1.0917
RxExperimental arm:crp1	0.5755	0.5443	1.7780
RxExperimental arm:tlr1	0.2522	0.3831	1.2869
RxExperimental arm:tmb_braf1	0.3657	0.3436	1.4415

4.4 Generalized Gamma

Parameter	Estimate	SE	exp(Est)
mu	3.0812	0.6541	21.7847
sigma	-0.5734	0.1448	0.5636
Q	0.2852	0.6756	1.3301
Age (per year)	0.0000	0.0089	1.0000
Sex (male vs female)	-0.1379	0.1651	0.8712
RxExperimental arm	-0.3974	0.3284	0.6721
crp1	0.0169	0.5005	1.0170
tlr1	0.9720	0.2843	2.6432
tmb_braf1	0.0874	0.2655	1.0914
RxExperimental arm:crp1	0.5741	0.5447	1.7756
RxExperimental arm:tlr1	0.2529	0.3833	1.2878
RxExperimental arm:tmb_braf1	0.3655	0.3436	1.4412

4.5 Gompertz

Parameter	Estimate	SE	exp(Est)
shape	0.0223	0.0052	1.0225
rate	-3.4252	1.1475	0.0325
Age (per year)	-0.0023	0.0166	0.9977
Sex (male vs female)	0.2329	0.3290	1.2623
RxExperimental arm	0.6939	0.5688	2.0016
crp1	0.3533	0.8147	1.4238
tlr1	-1.6909	0.5483	0.1844
tmb_braf1	-0.2949	0.4879	0.7446
RxExperimental arm:crp1	-1.3147	1.0200	0.2685
RxExperimental arm:tlr1	-0.2892	0.6760	0.7489
RxExperimental arm:tmb_braf1	-1.0307	0.7592	0.3568

4.6 Log-logistic

Parameter	Estimate	SE	exp(Est)
shape	1.0833	0.1187	2.9545
scale	3.2498	0.7145	25.7854
Age (per year)	-0.0034	0.0090	0.9966
Sex (male vs female)	-0.1807	0.1679	0.8347
RxExperimental arm	-0.4030	0.3671	0.6683
crp1	0.0133	0.4021	1.0134
tlr1	1.0309	0.3232	2.8036
tmb_braf1	0.0161	0.2996	1.0163
RxExperimental arm:crp1	0.5770	0.4699	1.7807
RxExperimental arm:tlr1	0.2448	0.3996	1.2774
RxExperimental arm:tmb_braf1	0.4047	0.3661	1.4988

4.7 Log-normal

Parameter	Estimate	SE	exp(Est)
meanlog	3.0492	0.6934	21.0982
sdlog	-0.5406	0.1029	0.5824
Age (per year)	-0.0008	0.0089	0.9992
Sex (male vs female)	-0.1413	0.1636	0.8683
RxExperimental arm	-0.3740	0.3369	0.6880
crp1	0.1227	0.4559	1.1305
tlr1	0.9463	0.2895	2.5761
tmb_braf1	0.0689	0.2661	1.0713
RxExperimental arm:crp1	0.4790	0.5195	1.6145
RxExperimental arm:tlr1	0.3088	0.3796	1.3618
RxExperimental arm:tmb_braf1	0.3505	0.3412	1.4198

4.8 Weibull (AFT)

Parameter	Estimate	SE	exp(Est)
shape	0.7437	0.1139	2.1038
scale	3.2164	0.5353	24.9377
Age (per year)	0.0015	0.0080	1.0015
Sex (male vs female)	-0.1076	0.1647	0.8980
RxExperimental arm	-0.4631	0.2751	0.6293
crp1	-0.1966	0.3853	0.8215
tlr1	0.9559	0.2475	2.6011
tmb_braf1	0.1323	0.2349	1.1414
RxExperimental arm:crp1	0.7180	0.4642	2.0504
RxExperimental arm:tlr1	0.1692	0.3137	1.1844
RxExperimental arm:tmb_braf1	0.4547	0.3454	1.5757

4.9 Weibull (PH)

Parameter	Estimate	SE	exp(Est)
shape	0.7437	0.1139	2.1038
scale	-6.7666	1.3964	0.0012
Age (per year)	-0.0032	0.0169	0.9968
Sex (male vs female)	0.2264	0.3463	1.2541
RxExperimental arm	0.9743	0.5936	2.6492
crp1	0.4137	0.8152	1.5123
tlr1	-2.0111	0.5566	0.1338
tmb_braf1	-0.2783	0.4940	0.7571
RxExperimental arm:crp1	-1.5106	0.9982	0.2208
RxExperimental arm:tlr1	-0.3560	0.6607	0.7005
RxExperimental arm:tmb_braf1	-0.9566	0.7384	0.3842

5 Interpretation Guide

5.1 Distribution-Specific Parameters

Different parametric distributions have different ancillary parameters:

Distribution	Ancillary Parameters
Exponential	rate
Weibull (AFT)	shape, scale
Weibull (PH)	shape, scale
Log-logistic	shape, scale
Log-normal	meanlog, sdlog
Gamma	shape, rate
Gompertz	shape, rate
Generalized Gamma	mu, sigma, Q
Generalized F	mu, sigma, Q, P

5.2 Coefficient Interpretation

- **Estimate**: The coefficient on the log scale (for AFT models) or log-hazard scale (for PH models)
- **SE**: Standard error of the coefficient estimate
- **exp(Est)**: Exponentiated coefficient
 - For AFT models: time ratio (values >1 indicate longer survival)
 - For PH models: hazard ratio (values <1 indicate lower hazard/better survival)

5.3 Biomarker Interaction Terms

The biomarker:Rx interaction terms indicate treatment effect modification:

- **crp1:RxExperimental** - Effect of experimental treatment in CRP-positive patients (relative to CRP-negative)
- **tlr1:RxExperimental** - Effect of experimental treatment in TLR-positive patients
- **tmb_braf1:RxExperimental** - Effect of experimental treatment in TMB_BRAF-positive patients

A significant interaction suggests the biomarker predicts differential treatment benefit.

6 Summary

This document presents the full set of parametric survival model coefficients for the METIMMOX-1 economic evaluation. The best-fitting models (selected by AIC) are:

Outcome	Best Distribution	AIC
Overall Survival (OS)	Gamma	622.56
Progression-Free Survival (PFS)	Gamma	453.43

All model coefficients are provided for transparency and to support sensitivity analyses exploring alternative distributional assumptions.

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